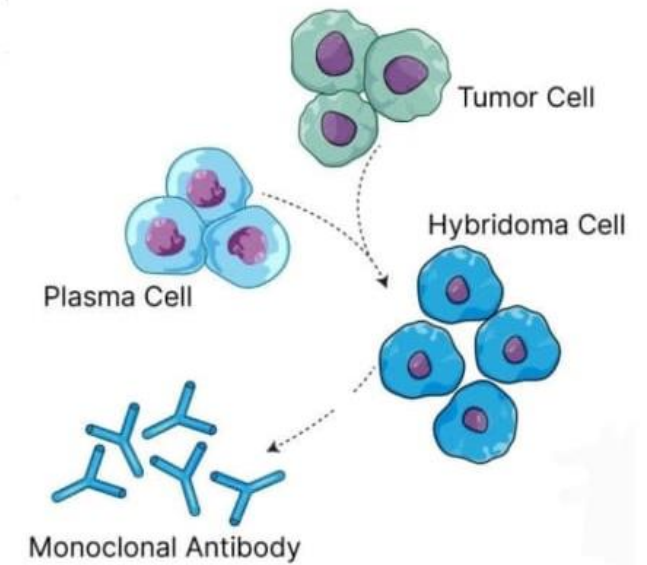


Fusion



- 
- Plasma cells (B cells) are responsible for Antibodies production against the Antigens that enter the body.
 - Plasma cells have short life span (few days) then Die.
 - **Hybridoma cells** : are cells result from fusion between plasma cells (Antibodies secreting cells) and Myeloma cells (immortal cancer B cells).
 - Hybridoma cells are immortal and secret Antibodies.



★ Myeloma cells which are used for fusion have specific Characteristics :

- HGPRT-
- Immortal
- Non producing and non secreting Anti-bodies.

★ Before fusion, myeloma cells :

- Isolate from log phase
- Cell density : not greater than 10^6 (one million).
- Cell viability more than 95%

- 
- Fusion between Myeloma and plasma cells achieved by :

- **Electroporation**

- **Chemicals:** poly ethylene glycol (PEG)

- PEG: is a polyether compound derived from petroleum with many applications, from industrial manufacturing to medicine.

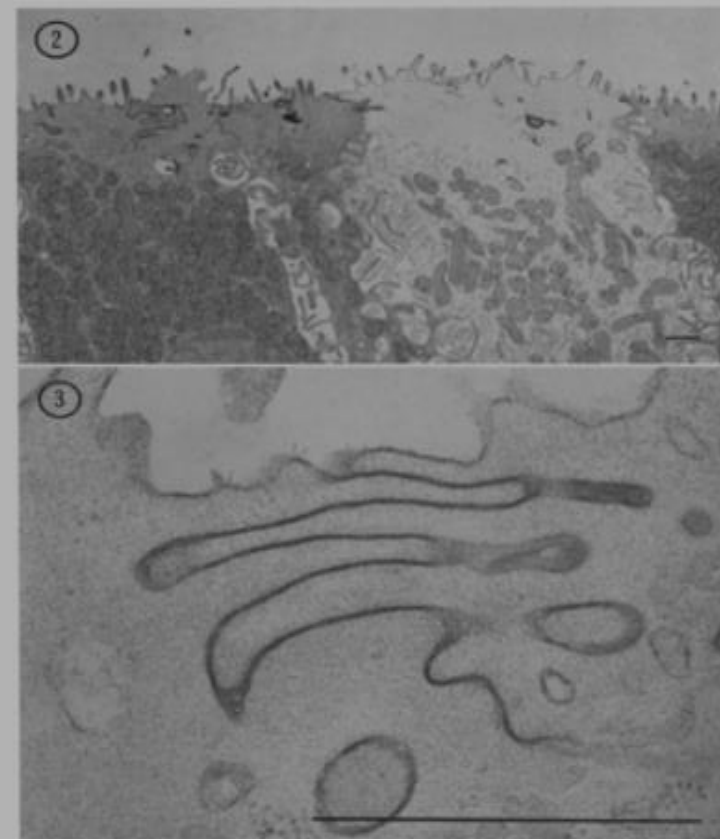
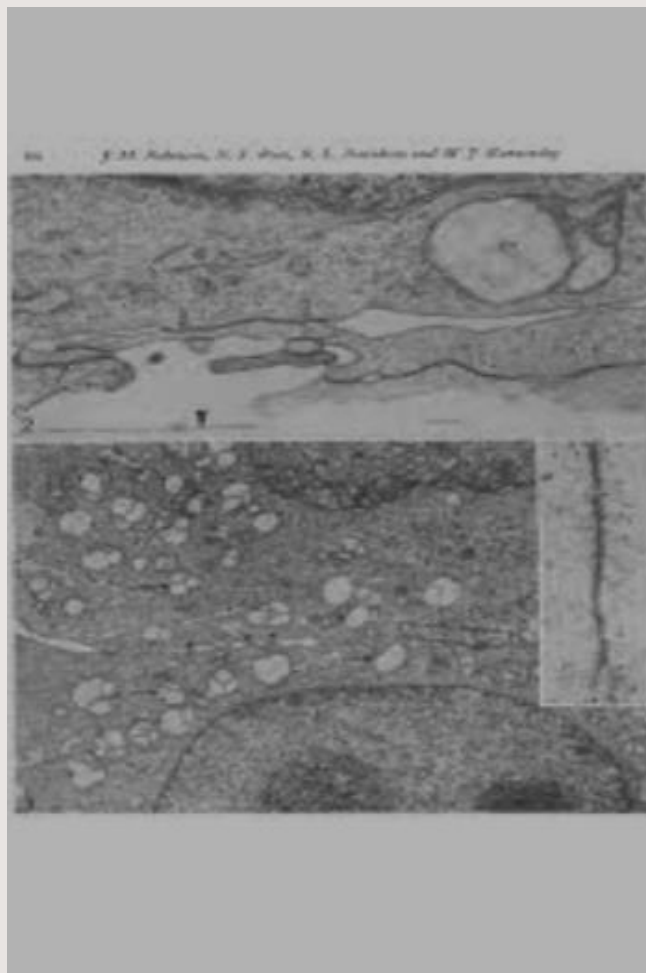
- The structure of PEG is commonly expressed as $\text{H}-(\text{O}-\text{CH}_2-\text{CH}_2)_n-\text{OH}$

- PEG induce rapid fusion
- Fusion detect by formation of pentalaminar membrane occur in 1 min after exposure to PEG



Advantages of using PEG :

1. unlikely to have specific interactions with biological chemicals (inert)
2. water-soluble polymer
3. low cytotoxicity for most cells
4. high fusion rate
5. ease of handling
6. reproducibility





For Fusion, PEG must has the following conditions :

- Molecular weight :1500-4000
- The concentration Of PEG: 40-50%
- PH: 8-8.2
- Time of exposure to PEG: 1-2 min
- Temp: 20-37oC
- Ratio of cells: 1:1-10:1 lymphocytes to Myeloma cells

- 
- The cells which isolate from spleen called splenocytes, splenocytes include : RBCs, macrophages, Dendritic cell, lymphocytes (B and T cells), and many different types of cells, so we have collection of different cells types, if you isolate one million cells from spleen its not only B cells, while the Myeloma cells culture is pure only contain

- 
- myeloma cells, typically for fusion ratio is 1-10 splenocytes :1 myeloma cells
 - For each myeloma cells we add approximately 10 splenocytes.
 - If you have one million Myeloma you should add 1-10 million of splenocytes to increase the Probability to Hybridoma formation.

DNA synthesis Pathways

Cells synthesis DNA by two pathways

1. De novo pathway: synthesize DNA from Scratch (from the beginning) from simple molecules such as formate and aspartate.

2. Salvage Pathway: recover bases and nucleotides from DNA that obtained from foods and use for Synthesis host Cell DNA

- 
- Salvage Pathway is important in some organs (brain and B.M) because some tissues cannot undergo de novo synthesis. The salvaged products can then be converted back into nucleotides.
 - Hypoxanthine Guanine phosphoribosyl transferase (HGPRT) is an important enzyme in salvage purine pathway metabolism and its deficiency is implicated in Lesch–Nyhan syndrome.
 - Mutation in Gene encode HGPRT • HGPRT Deficiency → Block salvage Pathways.



- Myeloma cells which used in hybridoma Technology are HGPRT- which mean only Synthesis DNA by De Novo Pathway

- After fusion achieved, five types of Cell exist in Culture :

- 1.Fused myeloma cells

- 2.Un fused myeloma

- 3.Un fused Plasma cells

- 4.Fused plasma cells

- 5.Hybridoma cells

Selection

- Select Hybridoma cells from unwanted cells.

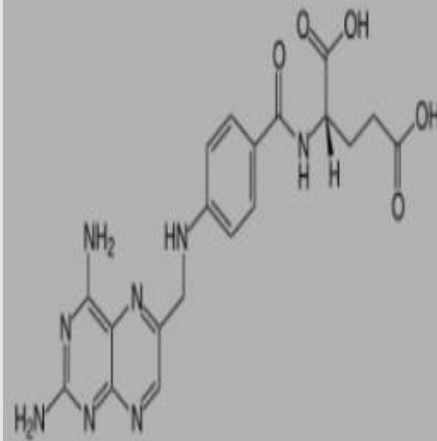
- Using selection media (HAT media)

HAT Medium (hypoxanthine-aminopterin-thymidine medium) is a selection medium for mammalian cell culture, which relies on the combination of:

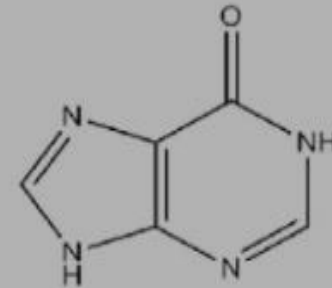
1-Hypoxanthine (a purine derivative) as precursor for purine synthesis

2-Aminopterin, a drug that acts as a powerful folate metabolism inhibitor by inhibiting dihydrofolate reductase.

3-Thymidine (a deoxynucleoside) which are intermediates in pyrimidine synthesis

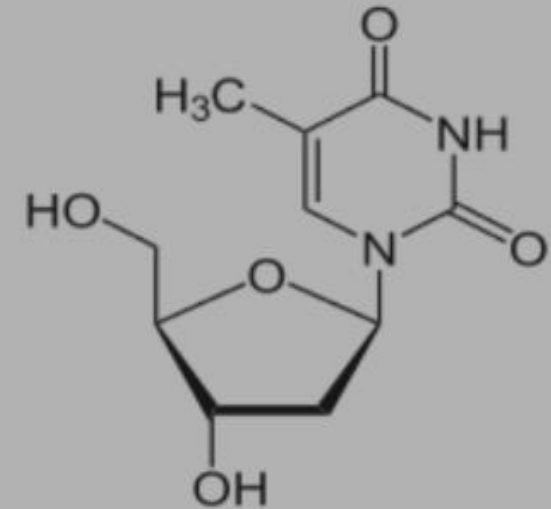


Aminopterin



Hypoxanthine
(Inosine)

Hypoxanthene



Thymidine

- 
- After fusion complete, the cells incubate overnight before culturing in HAT media, this because after fusion the cells are not ready for DNA synthesis it need some time to become stable and then start synthesis DNA.
 - Aminopterin will block De novo pathway which mean all cells present in HAT media have one option for DNA synthesis by using salvage Pathway.
 - Myeloma cells are HGPRT-, which mean cant synthesis DNA by salvage Pathways



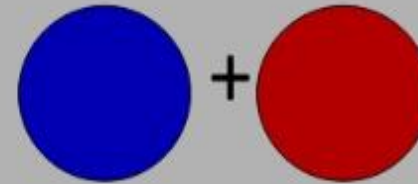
1-Myeloma cells will die, because They cant synthesis DNA by De Novo or salvage Pathway.

2- Un fused Plasma cells are HGPRT+ They can synthesis DNA by salvage Pathways, but Eventually will die because Plasma cells have short life span

3- hybridomas will survive because They have HGPRT+ (gain from plasma cells) which mean, they start DNA synthesis using salvage Pathway and also immortal(gain from cancer cells)

Possible Fusion Products

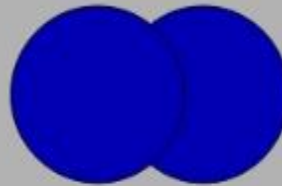
B Cells From
Immunized Animal



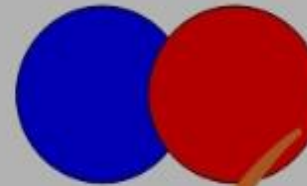
Myeloma HGPRT Deficient
And Ig Deficient



Senescence

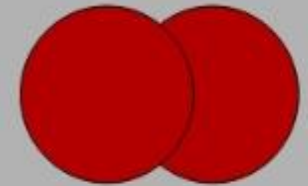


Senescence



Can Use Salvage
Pathway

No Senescence



HAT medium

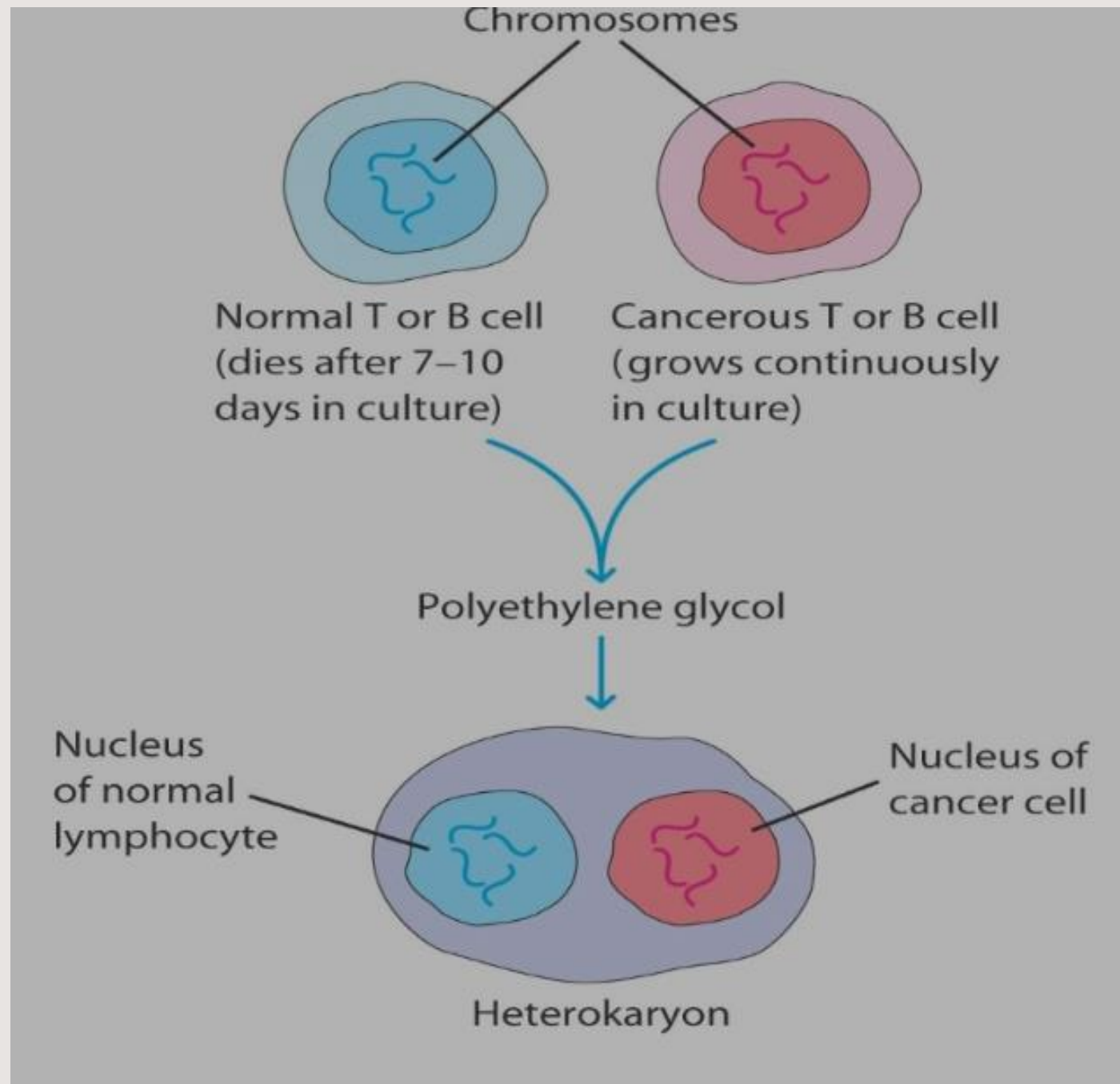


HAT
Medium



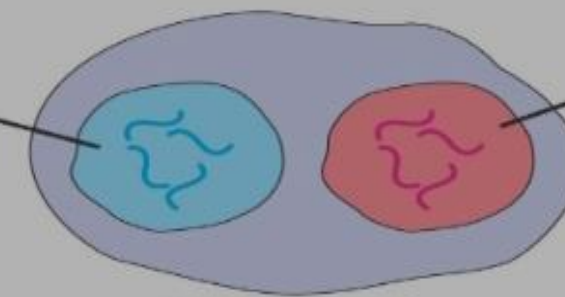
Heterokaryocyte

- Two somatic cells can be caused to fuse in vitro to produce a heterokaryocyte , which is a single hybrid cell containing 2 or more nuclei.
- The first mitotic division after fusion will lead to a population of daughter cells that carry both sets of parent cell chromosomes in the same nucleus.
- Often, at subsequent divisions chromosomes are lost in ones and twos until a new cell line is formed carrying some chromosomes from each parent.



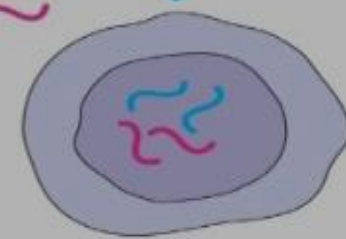
Nucleus
of normal
lymphocyte

Nucleus of
cancer cell



Heterokaryon

Random chromosomal loss



Hybridoma

(expresses some normal B-cell or T-cell genes but
grows indefinitely like a cancer cell)

Fusion protocol

- 1- Add myeloma cells and B cells together
2. Centrifuge mixed cells 900-1000 rpm 5 min.
3. Slowly add 1 ml PEG to the pelleted cells.
(Do at room temperature)

Add of PEG drop by drop not at one time.

Dispense PEG along sides of tube. After that, mix by Swirling tube.

4. Incubate for 1 minute at 37°C

After incubation, PEG dilute by Complete media to stop the activity of PEG.



5. Distribute cell mixtures in 96-well plate

6. Top up medium in each well

7. Leave plate in 5% CO₂, 37°C incubator overnight before adding HAT medium

- The FCS must be pretested for its ability to support clonal growth
- After addition of PEG all handling of the cells should be as gentle as possible to minimize dispersal of forming hybrids.